

**Get Light of 18W LED Tube in Just 12W Only!**

**Now Switch To**

**4-foot T-8**

**ULTRA HIGH EFFICIENCY  
PowerLED Tube Light**

(Power Consumption 12W only; SAME light output)

ULTRA HIGH EFFICIENCY

**33%**  
Lower Power  
Consuming

**MOST  
EFFICIENT POWER  
RATING LED  
TUBE LIGHTS IN  
THE MARKET!**

While most comparative competitive LED tubes achieve a lumen output of 1600 lumens at 18W approx, the BINAY ULTRA HIGH EFFICIENCY QUADRA-12-ES PowerLED Tube Light is designed for 2000 source lumens at a 33% lower power consumption of 12W, thus achieving considerable power savings (apart from much longer maintenance free life). The light output performance is equivalent to that of the 1200mm T-8 fluorescent tube.

**5 YEARS Warranty WITH AN ESTIMATED LIFE OF 12-15 YEARS**



**Binay Opto Electronics Pvt. Ltd.**

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**POWERING LED TECHNOLOGIES WORLDWIDE SINCE 1983**

## BINAY Multichip LED Lamps

High reliability direct-retrofit LED bulbs for Control and Instrumentation applications



Ba9s bayonet pin cap



E-10 screw cap



Midget Flange cap



Midget Groove cap



5530 and 6844 slide base



Wedge base

Simple retrofits for existing filament bulb holders, in applications such as annunciator window (fascia) backlighting and illuminated pushbutton switches, where the reliability of indication is important or where lamp replacement is difficult or expensive

- Long life (typically 100,000 hours)
- Low power consumption;
- Low heat generation
- High brightness and wide-angle light emission
- Shock-proof and vibration resistant
- High reliability of indication
- One-time Expense



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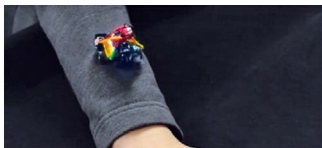
## Ultra-low power transistors that could function for years without a battery

A newly-developed form of transistor opens up a range of new electronic applications including wearable or implantable devices by drastically reducing the amount of power used. Devices based on this type of ultra-low power transistor, developed by engineers at University of Cambridge, could function for months or even years without a battery, by scavenging energy from its environment.

Using a principle similar to a computer in sleep mode, this new transistor harnesses a tiny leakage of electrical current, known as a near-off-state current, for its operations. This leak, like water dripping from a faulty tap, is a characteristic of all transistors. But this is the first time that it has been effectively captured and used functionally. The results open up new avenues for system design for the Internet of Things, in which most things we interact with every day are connected to the Internet.

## Tiny robots roam around the body

Researchers from Massachusetts Institute of Technology and Stanford University have developed miniature wearable robots, Rovables, that can crawl down and around unmodified garments; you do not need to wear any special garment to make these crawl on you.



Rovables come with low-power localisation systems that allow these to move autonomously (Image courtesy: <http://rb.ru>)

With the help of magnetic gripping wheels on either side of clothing, Rovables can scale garments. Sensors and artificial intelligence help these to be partially autonomous. A little bigger than a ring box, these tiny wearable robots can house wireless communications, microcontrollers and a battery that can stay charged for 45 minutes at full capacity before needing to be charged again. However, researchers say that battery life should extend for much longer if the robots are not moving all the time.

These robots can be quite useful. For example, when there is a new email, these can tap on your body, or roll up your sleeve if your body temperature increases. Also, these can form a light on your back during an evening bike ride or assemble on your arm to create a display and play video.